

On the Treatment of Gauge Zero Modes in Yang-Mills Theory.

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In the computation of the instanton contribution to the functional integral^(1,2) there is a very delicate point: the treatment of zero modes associated with the invariances of the theory. Those related with translations and dilatations can be handled following the standard collective-co-ordinate method⁽³⁾.

On the other hand, gauge zero modes in the background gauge⁽⁴⁾, which is the natural one⁽⁵⁾, require special care. In a remarkable paper⁽¹⁾, 't Hooft developed a procedure to take into account correctly this problem, showing that:

1) There are three anomalous zero modes related to global isospin rotations, which must not be replaced by a collective co-ordinate.

2) The redundant eigenmodes can be fixed by an additional component in the gauge-fixing term.

The method used by 't Hooft to handle these difficulties is extremely subtle. It is the purpose of this note to clarify the problem of gauge zero modes, stressing its geometrical meaning. For that, we will use the mathematical framework settled by SINGER⁽⁶⁾ in the context of the Gribov problem. We will then see that a rigorous treatment shows that the 't Hooft procedure is the final link of a chain of restrictions on gauge degrees of freedom, that one must impose in order to render the functional integral finite: with the usual background gauge condition^(4,5), a class of zero modes associated with local gauge invariance is removed. This class is bigger than the one eliminated with the orthogonality (or generalized Coulomb) condition⁽⁶⁾. But there are gauge degrees of

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freedom, related to global gauge rotations, which escape from the geometrical prescription and which must be fixed by additional conditions.

Let us begin by stating 't Hooft results: when one computes vacuum-vacuum amplitudes expanding the gauge field A_μ , about the instanton configuration A_μ^{cl} ,

$$(1) \quad A_\mu = A_\mu^{\text{cl}} + a_\mu,$$

the functional integral, Z , takes the form

$$(2) \quad Z = \int \mathcal{D}A_\mu \exp[-S[A]] = \int \mathcal{D}a_\mu \exp[-S^{\text{cl}} - S^{\text{II}}],$$

where the quadratic part of the action, S^{II} , is given by

$$(3) \quad S^{\text{II}} = \frac{1}{2} \int a_\mu \mathcal{L}_{\mu\nu}^{\text{II}} a_\nu dx.$$

In order to perform the functional integration, this quadratic form must be diagonalized. It is then necessary to obtain eigenvalues and eigenfunctions of the equation

$$(4) \quad \mathcal{L}_{\mu\nu}[A^{\text{cl}}] a_\nu^{(n)} = \lambda^{(n)} a_\mu^{(n)}.$$

Now, since the action is gauge invariant, an infinitesimal gauge transformation of the form

$$(5) \quad D_\mu^{\text{cl}} \psi(x),$$

where

$$(6) \quad D_\mu^{\text{cl}} = \partial_\mu + [A_\mu^{\text{cl}}]$$

leaves $S(A)$ invariant and is then a zero mode of eq. (4). Since S^{II} is self-conjugated, eigenfunctions with different eigenvalues are orthogonal. In particular,

$$(7) \quad \text{tr} \int a_\mu^{(n)} \cdot D_\mu^{\text{cl}} \psi dx = 0.$$

Now, if ψ is well behaved at infinity, as to allow integration by parts disregarding surface terms, eq. (7) implies

$$(8) \quad D_\mu^{\text{cl}} a_\mu^{(n)} = 0,$$

$\forall n$ such that $\lambda^{(n)} \neq 0$.

This is the background gauge condition (4), which then turns to be the «natural» one (5). By fixing the gauge through the δ -function:

$$(9) \quad \delta(D_\mu^{\text{cl}} A_\mu),$$

incorporated to Z with the corresponding Faddeev-Popov determinant:

$$(10) \quad \Delta = \det(D_\mu D_\mu^{\text{cl}}),$$

one eliminates gauge degrees of freedom associated to zero modes of the form (5).

Now, apart from the remaining translation and dilatation zero modes, there are three spurious gauge zero modes discovered by 't Hooft ⁽¹⁾:

$$(11) \quad a_{\mu}^{(0)}(b) = D_{\mu}^{cl} \psi(b), \quad b = 1, 2, 3,$$

which satisfy

$$(12) \quad D_{\mu}^{cl} a_{\mu}^{(0)}(b) = 0,$$

with

$$(13) \quad a_{\mu}(b) = \frac{\sigma^a}{i} \eta_{ak\mu} \bar{\eta}_{bk\lambda} \frac{x^{\lambda}}{(1+x^2)^2},$$

$$(14) \quad \psi^a(b) = \eta_{ak\mu} \eta_{bk\lambda} \frac{x^{\mu} x^{\lambda}}{1+x^2}.$$

The norm of $a_{\mu}^{(0)}$ is

$$(15) \quad \mathcal{N} = -\text{tr} \int a_{\mu}^{(0)} a_{\mu}^{(0)} dx = 2\pi^2.$$

These zero modes, indirectly related to global isospin rotations, have the following characteristics:

- 1) They are a pure gauge artifact.
- 2) When written in the form (11), the corresponding ψ 's take at infinity a value which depends on the direction infinity approached. The norm of $\psi(b)$, $-\text{tr} \int \psi \psi dx$ is infinite.
- 3) They are not eliminated by the fixing gauge procedure described above.
- 4) The zero eigenvalues contribute both to the vector and ghost determinants, since $\psi(b)$ are zero modes of the Faddeev-Popov operator.

Notice that 2) implies that the integration by parts we performed to obtain eq. (8), leaves in this case a surface term.

The relation of these zero modes with isospin orientation can be easily seen in the singular gauge ⁽²⁾.

In order to handle these spurious zero modes, 't Hooft changed the background gauge condition, in order to compute their contribution to the functional integral and extract it from the final result.

We will now interpretate these facts using the geometrical framework settled by SINGER ⁽⁶⁾. For definiteness we will consider in expression (1) that A_{μ}^{cl} corresponds to a charge-one instanton configuration. That is,

$$(16) \quad \left\{ \begin{array}{l} A_{\mu}^{cl} = f(x) g^{-1} \partial_{\mu} g, \\ f = \frac{x^2}{1+x^2} \end{array} \right.$$

with g belonging to the charge-one sector:

$$(17) \quad \begin{cases} g: \mathbf{R}^4 - \{0\} \rightarrow SU_2, \\ g = \frac{x_4 + i\boldsymbol{\sigma} \cdot \mathbf{x}}{x^2}. \end{cases}$$

We denote by $P(\mathbf{R}^4, SU_2)$ the trivial principal fiber bundle over $\mathbf{R}^4$ with structural group SU_2 . Although one may consider that the instanton configuration defines a connection on a nontrivial principal fiber bundle over S^4 , constructed through the transition function g , eq. (17), in the context of 't Hooft work, the natural base space is $\mathbf{R}^4$. The norm defined by eq. (15) is the usual one in $\mathbf{R}^4$. Had we worked with configurations projected from S^4 back to $\mathbf{R}^4$, the correct norms would be different (⁷).

Following Singer (⁶), we define $\mathcal{B} = \{A/A \text{ is an irreducible connection for } \mathcal{P}\}$. $\mathcal{B}$ is an infinite dimensional manifold. We call $\mathcal{G}$ the group of gauge transformations which are the identity at infinity.

Now, let us define A^1 as the set of equivariant one-forms on $\mathbf{R}^4$, with values in $\mathcal{A}$, the Lie Algebra of SU_2 . That is, $A^1 = \{a/a \text{ is a one form with values in the adjoint bundle of } \mathcal{P} \text{ with fiber } \mathcal{A}\}$. A^1 is isomorphic to the tangent space $T_A(\mathcal{B})$, $\forall A \in \mathcal{B}$.

Analogously, we define A^0 as the set of equivariant functions ψ with values in $\mathcal{A}$, $\psi \rightarrow 0$ at infinity. A^0 is isomorphic to the Lie Algebra of the group $\mathcal{G}$.

Then, the tangent space of the orbit by A is given by the image of the covariant differential

$$(18) \quad D_A: A^0 \rightarrow A^1.$$

We will define, in a natural way, inner products for A^0 and A^1 :

$$(19) \quad (\psi, \varphi) = -\text{tr} \int dx \psi(x) \varphi(x),$$

$$(20) \quad (a, b) = -\text{tr} \int dx a_\mu(x) b_\mu(x),$$

where $\psi, \varphi \in A^0$ and $a = a_\mu dx^\mu$, $b = b_\mu dx^\mu \in A^1$.

Note that expression (20) gives the adequate inner product to handle the functional integration in the form explained at the beginning. Equation (19) gives the natural inner product for ψ , considering that A^0 is the space on which the Faddeev-Popov operator acts (⁸).

We now restrict A^0 and A^1 requiring finiteness of the norms defined from eqs. (19)-(20). In fact, we will consider in the following the (usual) more restrictive condition:

$$\psi \underset{x^2 \rightarrow \infty}{\sim} 1/x^{2+\varepsilon}$$

and the same for a_μ .

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(⁸) O. BABELON and C. M. VIALLET: *Phys. Lett. B*, **85**, 246 (1979).

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With these restrictions, the gauge condition (generalized Coulomb gauge) defined by Singer ⁽⁶⁾, does not correspond exactly to the condition (eq. (8))

$$(21) \quad D_{\mu}^{cl} a_{\mu} = 0.$$

Actually, SINGER fixes the gauge by taking the orthogonal complement to the tangent at the orbit, that is, the kernel of D_A^* , the adjoint of D_A . On the other hand, condition (21) is more restrictive, since it implies orthogonality to a bigger set. In fact, any $a_{\mu} = D_{\mu}^{cl} \psi \in A'$ with ψ ($\psi \notin A^0$), behaving at infinity as

$$(22) \quad \psi \xrightarrow{x \rightarrow \infty} 1/x^{1+\varepsilon}, \quad 0 < \varepsilon \leq 1$$

is not tangent to the orbit, but it lies in the orthogonal complement to the set defined by condition (21).

Hence, for the purpose of functional integration, it is convenient to fix the gauge with the usual condition (21), since in this way one also removes gauge zero modes of the form $D_{\mu}^{cl} \psi$ with ψ satisfying condition (22). Otherwise, by using the orthogonal complement to the « well defined » orbit, one will face the problem of zero modes coming from the projection of these a_{μ} 's.

Apart from these zero modes, there are also the spurious ones, eqs. (11)-(14), coming from ψ 's not belonging to A^0 . Neither are they eliminated by fixing the gauge with condition (21), nor, *a fortiori*, by Singer's generalized Coulomb gauge. To handle correctly these three spurious zero modes, one must eliminate the three dimensional subspace Z_3 , generated by them.

This has been done by 't Hooft, who changed condition (21) by adding a term that fixes the remaining degrees of freedom. In this way, the zero eigenvalues are replaced by nonzero constants (Of course, since (ψ, ψ) is not finite, one must work in a finite space-time volume). One can then compute the contribution of Z_3 to the functional integral in order to extract it from the final result.

We see that fixing the gauge by an orthogonality condition leaves redundant degrees of freedom which come from ψ 's not belonging to A^0 . Since, for computational purposes, one must restrict oneself to space A^0 , the redundant degrees of freedom must be fixed by additional constants.

We note that, since the set defined by eq. (21) is smaller than the orthogonal complement to the orbit, difficulties can arise in finding the representatives, in the background gauge, of certain configurations close to the instanton ⁽⁹⁾.

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